



FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

IT0049 (WEB SYSTEM TECHNOLOGIES)

TECHNICAL FORMATIVE ASSESSMENT

3

MAKING IT EDITABLE: FORMS, VALIDATION, AND FILE UPLOAD

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PROGRAM OUTCOME/S (PO) ADDRESSED BY THE LABORATORY EXERCISE

- Design, implement and evaluate computer-based systems or applications to meet desired needs and requirements. [PO: C]

COURSE LEARNING OUTCOME/S (CLO) ADDRESSED BY THE LABORATORY EXERCISE

- Apply advanced web development principles in creating, debugging, validating, and securing database-backed web applications. [CLO: 2]

INTENDED LEARNING OUTCOME/S (ILO) OF THE LABORATORY EXERCISE

At the end of this exercise, students must be able to:

- Build and validate HTML forms for creating new records.
- Redisplay a form with validation errors and the user's previously entered values.
- Implement an edit and update workflow for existing records.
- Accept, validate, prepare, and store an uploaded image file, saving only its filename in the database.

BACKGROUND INFORMATION

A read-only application is only half useful. This activity adds the ability to create and edit records through validated forms — rejecting bad input instead of trusting it — and introduces file upload, a common requirement for anything involving user avatars or product images. CodeIgniter's validation rules, `getFile()` method, and Image service handle these concerns without changing the main upload workflow.



GRADING SYSTEM / RUBRIC

CRITERIA	Description
Functionality & Requirements Completeness (40 points max)	All required pages and features are present and work as specified in the activity instructions.
Code Structure & Organization (25 points max)	Controllers, models, and views are separated correctly and follow the framework's conventions; code is readable and sensibly named.
Form Validation & File Handling (20 points max)	Forms reject invalid input with clear error messages and preserve the user's entries on redisplay; the uploaded file is validated by type and size, prepared for display, stored safely, and only its filename is saved to the database.
Documentation & Submission Quality (15 points max)	The GitHub repository is organized and complete, the README explains how to set up and run the project, and the hosted link is live and matches the submitted code.

LABORATORY ACTIVITY

Extend the POS application from TFA2 with the ability to create and edit customer and user accounts, and add avatar upload with display-ready image preparation for user accounts.

1. Add a New Customer form (/customers/new) that validates required fields (full name required, email required and valid) before inserting a record.
2. Add a New User form (/users/new) that validates required fields (username required and unique, full name required) before inserting a record.
3. Add edit pages for both customer and user accounts that pre-fill the form with the existing record's data and update it on submit.
4. Add an avatar column to the users table. On the user edit form, add a file input for a profile picture; validate that the uploaded file is a JPG or PNG no larger than 2MB, prepare a display-ready copy or thumbnail, store it in a public uploads folder, and save only the filename in the avatar column.
5. Display each user's prepared avatar or thumbnail on the User Accounts listing page (fall back to a placeholder image if none was uploaded).



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Submission Requirements:

- A link to a GitHub repository containing your raw project files, including your database export.
- A link to your hosted, working version of the application.

REFERENCES

- CodeIgniter User Guide. Validation.
<https://codeigniter4.github.io/userguide/libraries/validation.html>
- CodeIgniter User Guide. File Uploads.
https://codeigniter4.github.io/userguide/libraries/uploaded_files.html



```
[mysql> USE pos_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
[mysql> DESCRIBE users;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra           |
+-----+-----+-----+-----+-----+-----+
| id         | int           | NO   | PRI | NULL    | auto_increment |
| username   | varchar(50)   | NO   | UNI | NULL    |                 |
| full_name  | varchar(100)  | NO   |     | NULL    |                 |
| avatar     | varchar(255)  | YES  |     | NULL    |                 |
| created_at | datetime      | NO   |     | NULL    |                 |
+-----+-----+-----+-----+-----+-----+

5 rows in set (0.010 sec)

mysql>
```

Figure 1. Users table updated with the avatar column

```
.env UserModel.php M CustomerModel.php M X
pos-system > app > Models > CustomerModel.php > ...
1  <?php
2
3  namespace App\Models;
4
5  use CodeIgniter\Model;
6
7  5 references | 0 implementations
8  class CustomerModel extends Model
9  {
10     0 references
11     protected $table = 'customers';
12     0 references
13     protected $primaryKey = 'id';
14     0 references
15     protected $returnType = 'array';
16
17     0 references
18     protected $allowedFields = [
19         'full_name',
20         'email',
21         'phone',
22         'created_at',
23     ];
24 }
```

Figure 2. CustomerModel used for customer create and update operations



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```
.env UserModel.php M CustomerModel.php M
pos-system > app > Models > UserModel.php > ...
1  <?php
2
3  namespace App\Models;
4
5  use CodeIgniter\Model;
6
7  5 references | 0 implementations
8  class UserModel extends Model
9  {
10     0 references
11     protected $table = 'users';
12     0 references
13     protected $primaryKey = 'id';
14     0 references
15     protected $returnType = 'array';
16
17     0 references
18     protected $allowedFields = [
19         'username',
20         'full_name',
21         'avatar',
22         'created_at',
23     ];
24 }
```

Figure 3. UserModel configured to store the avatar filename

POS Management / New Customer

CUSTOMER ACCOUNTS

New Customer

Add a customer using the required contact fields.

CUSTOMER DETAILS

Create customer account

Full name *

Enter the customer's complete name.

Email address *

Use a valid email address such as name@example.com.

Phone number

Optional. Maximum of 20 characters.

Create customer Cancel

Figure 4. New Customer form



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CUSTOMER DETAILS

Create customer account

Please correct the highlighted fields.

The Email address field must contain a valid email address.

Full name *

Juan Dela Cruz

Enter the customer's complete name.

Email address *

hello

Use a valid email address such as name@example.com.

The Email address field must contain a valid email address.

Phone number

09291234566

Optional. Maximum of 20 characters.

Create customer

Cancel

Figure 5. Customer form rejecting invalid input while preserving entered values

Customer created successfully.

ACCOUNT DIRECTORY

Customer Accounts

6
TOTAL RECORDS

Contact details and account metadata for customers stored in the POS database.

DATABASE DIRECTORY

Customer records

customers table

New Customer

ID	FULL NAME	EMAIL	PHONE	CREATED AT	ACTIONS
#1	Angela Reyes	angela.reyes@example.com	+63 917 245 8103	Sep 1, 2026 · 9:15 AM	Edit
#2	Marco Villanueva	marco.villanueva@example.com	+63 918 632 4751	Sep 2, 2026 · 10:30 AM	Edit
#3	Sofia Mendoza	sofia.mendoza@example.com	+63 905 381 9264	Sep 3, 2026 · 11:45 AM	Edit
#4	Daniel Lim	daniel.lim@example.com	+63 922 714 5386	Sep 4, 2026 · 1:20 PM	Edit
#5	Bianca Santos	bianca.santos@example.com	+63 916 849 2075	Sep 5, 2026 · 3:10 PM	Edit
#6	Juan Dela Cruz	Juandelacruz@gmail.com	09291234566	Oct 3, 2026 · 12:48 AM	Edit

Figure 6. Newly created customer displayed in Customer Accounts



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CUSTOMER ACCOUNTS

Edit Customer

Update the selected customer's contact details.

CUSTOMER #6
Update customer account

Full name *

Enter the customer's complete name.

Email address *

Use a valid email address such as name@example.com.

Phone number

Optional. Maximum of 20 characters.

Figure 7. Customer Edit form pre-filled with existing record data

Customer updated successfully.

ACCOUNT DIRECTORY

Customer Accounts

6
TOTAL RECORDS

Contact details and account metadata for customers stored in the POS database.

DATABASE DIRECTORY
Customer records

ID	FULL NAME	EMAIL	PHONE	CREATED AT	ACTIONS
#1	Angela Reyes	angela.reyes@example.com	+63 917 245 8103	Sep 1, 2026 - 9:15 AM	Edit
#2	Marco Villanueva	marco.villanueva@example.com	+63 918 632 4751	Sep 2, 2026 - 10:30 AM	Edit
#3	Sofia Mendoza	sofia.mendoza@example.com	+63 905 381 9264	Sep 3, 2026 - 11:45 AM	Edit
#4	Daniel Lim	daniel.lim@example.com	+63 922 714 5386	Sep 4, 2026 - 1:20 PM	Edit
#5	Bianca Santos	bianca.santos@example.com	+63 916 849 2075	Sep 5, 2026 - 3:10 PM	Edit
#6	Juan Cruz	Juandelacruz@gmail.com	09291234566	Oct 3, 2026 - 12:48 AM	Edit

Figure 8. Updated customer record displayed successfully



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USER ACCOUNTS

New User

Create a staff user with a unique username.

USER DETAILS

Create user account

Username *

Names must be unique.

Full name *

Enter the staff user's complete name.

Create user **Cancel**

Figure 9. New User form

USER DETAILS

Create user account

Please correct the highlighted fields.

- That username is already in use.

Username *

Names must be unique.

That username is already in use.

Full name *

Enter the staff user's complete name.

Create user **Cancel**

Figure 10. Duplicate username rejected by server-side validation



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User created successfully.

STAFF DIRECTORY

User Accounts

Identity and account metadata for staff users stored in the POS database.

6
TOTAL RECORDS

DATABASE DIRECTORY

Staff user records

users table

New User

ID	AVATAR	USERNAME	FULL NAME	CREATED AT	ACTIONS
#1		acruz	Andrea Cruz	Sep 1, 2026 - 8:00 AM	Edit
#2		jnavarro	Joshua Navarro	Sep 2, 2026 - 8:30 AM	Edit
#3		mpascual	Mikaela Pascual	Sep 3, 2026 - 9:00 AM	Edit
#4		rgonzales	Rafael Gonzales	Sep 4, 2026 - 9:30 AM	Edit
#5		tgarcia	Trisha Garcia	Sep 5, 2026 - 10:00 AM	Edit
#6		jcruz	Juan dela cruz	Oct 3, 2026 - 1:04 AM	Edit

Figure 11. Newly created user displayed in User Accounts

Edit User

Update account details and optionally replace the profile picture.

USER #6

Update user account

Username *

jcruz

Names must be unique.

Full name *

juan dela cruz

Enter the staff user's complete name.

Profile picture

The current image remains when no replacement is selected.

Choose File

No file chosen

Optional JPG or PNG only. Maximum size: 2 MB.

Save changes

Cancel

Figure 12. User Edit form with profile-picture upload



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#5		tgarcia	Trisha Garcia	Sep 5, 2026 · 10:00 AM	Edit
#6		jcruz	juan dela cruz	Oct 3, 2026 · 1:04 AM	Edit

Figure 13. User Accounts page displaying the uploaded avatar